1. Determination of the value of *x* in the hydrated copper(II) sulfate, $CuSO_4.xH_2O$

FA 1 is 0.150 mol dm^{-3} sodium thiosulfate, $Na_2S_2O_3$.

FA 2 is dilute sulfuric acid.

FA 3 is 1.00 mol dm^{-3} potassium iodide, KI.

FA 4 is a solution made by dissolving 32.5 g of $CuSO_4.\mathbf{x}H_2O$ in 1.00 dm^3 of solution.

starch indicator

In this experiment you will perform titrations to determine the value of **x** in the formula for hydrated copper(II) sulfate, $CuSO_4.\mathbf{x}H_2O$, and its concentration in **FA 4**. You will first react a solution of Cu^{2+} ions with excess iodide ions, I^-. This reaction produces iodine as shown in equation 1.

equation 1 $$2Cu^{2+}(aq) + 4I^-(aq) \rightarrow 2CuI(s) + I_2(aq)$$

I_2 has relatively low solubility in water. However, in the presence of excess I^-, the soluble triiodide ion, I_3^- is formed, as shown in equation 2.

equation 2 $$I_2(aq) + I^-(aq) \rightarrow I_3^-(aq)$$

The I_3^- ions formed may be titrated against a standard solution of $Na_2S_2O_3$ as shown in equation 3.

equation 3 $$I_3^-(aq) + 2S_2O_3^{2-}(aq) \rightarrow 3I^-(aq) + S_4O_6^{2-}(aq)$$

(a) Procedure

1. Fill the burette with **FA 1**.
2. Pipette 25.0 cm^3 of **FA 4** into a 250 cm^3 conical flask.
3. Use a 10 cm^3 measuring cylinder to transfer 10.0 cm^3 of **FA 2** to the same conical flask.
4. Use a second 10 cm^3 measuring cylinder to transfer 10.0 cm^3 of **FA 3** to the same conical flask. A white precipitate forms in a brown solution.
5. Add **FA 1** from the burette into this flask. Near the end-point, when the brown solution becomes pale, add about 10 drops of starch indicator.
6. Continue adding **FA 1** slowly. The end-point is reached when the **solution** first becomes colourless. The white precipitate remains.
7. Repeat steps 2 to 6 and step until consistent results are obtained.

Results

M1

M2

M3

M4

[5] M5

(b) From your titrations, obtain a suitable volume of **FA 1** to be used in your calculations.

Show clearly how you obtained this volume

Volume of **FA 1** = …….…… cm^3 **[1]** M6

1 **(c)** **(i)** Calculate the number of moles of $S_2O_3^{2-}$ in the volume of **FA 1** in **(b)**.

Number of moles of $S_2O_3^{2-}$ in **FA 1** = ……..…….. **[1]** M7

(ii) Calculate the number of moles of copper(II) ions in 25.0 cm^3 of **FA 4**.

Number of moles of Cu^{2+} in **FA 4** = ……..…….. **[1]** M8

(iii) Calculate the value of **x** in $CuSO_4.\mathbf{x}H_2O$.

[A_r: Cu, 63.5; S, 32.1; O, 16.0; H, 1.0]

x = …….....…….. **[1]** M9

1 **(c)** **(iv)** Blue vitriol solution is the concentrated form of **FA 4**.

FA 4 was prepared by diluting 10.0 cm^3 of blue vitriol solution to 250 cm^3 in a volumetric flask using deionised water.

Hence or otherwise, determine the concentration of Cu^{2+} in blue vitriol solution in mol dm^{-3}.

Concentration of Cu^{2+} in blue vitriol solution: ..... mol dm^{-3} **[1]** M10

(d) A student performed a titration and obtained a titre volume of 30.50 cm^3. Calculate the maximum percentage error in the student's titre volume.

Maximum percentage error = ……..

[1] M11

1 **(e)** A student suggests that the experiment could be made more accurate if the volume of **FA 3** was measured using a burette.

Suggest a reason why this change would not improve the accuracy of the experiment.

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……..... **[1]** M12

(f) A laboratory technician accidentally prepared **FA 4** by dissolving 32.5 g of hydrated copper(II) sulfate in 250 cm³ of water. Suggest how this mistake would affect the volume of **FA 1** required in the titration and the calculated value of **x** in $CuSO_4.\mathbf{x}H_2O$.

Explain your answer.

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……... **[2]** M13 M14

(g) The formation of the white CuI precipitate can make it harder to determine the end–point of the titration. Suggest another method to measure the amount of iodine produced without using titration.

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……... **[1]** M15

[Total: 15]